Course architecture

Advanced Statistics 2027-1 / Probabilistic Machine Learning · Course-level intellectual and curricular authority

This document defines the **intellectual purpose and planned progression** of the course. It is the primary reference when selecting lecture scopes, weighing changes across lectures, and interpreting the relationship between mathematical results and applications. The student syllabus is a concise description of this architecture; approved lecture scopes determine their own detailed teaching commitments. Actual delivery and source coverage are recorded in the material ledger. These authorities should be reconciled when they disagree, not silently overwritten.

What the course is about

Understanding probabilistic machine learning is understanding **the probabalistic formulation of machine language problems**, **how different machine language problems are expressed in this machinery**, and **why specific classes of learning problems apply different computational approaches to the same machinery**. The goal is to achieve a coherent and integrated view of how the structured data, the questions being asked, and the computational tools are all related through the probabalistic formulation. This will not be a catalogue of models and it is not a progression from one approximation algorithm to the next. Each mathematical advance is motivated by a kind of task and available experience for which our existing models, performance criteria, or computations are insufficient.

From learning problems to mathematical formulations

A **learning task** specifies what a system should accomplish; its **experience** specifies the data, feedback, or interaction available. Its success must be defined by a **performance criterion**. The same dataset may support prediction, generation, discovery, or action; data alone do not specify the learning problem.

The mathematical formulation has **two distinct, related objects**. A probabilistic model describes the relevant observed and unobserved variables, their dependencies, and our uncertainty. A performance functional defines what counts as a good prediction, representation, decision, or learned distribution—often by taking an expectation with respect to a relevant distribution. For example, a predictive action a may be evaluated by

$$\mathcal{R}(a) = \int L(a, y) p(y) \, \mathrm{d}y.$$

The distribution under which we evaluate actual performance need not be the same as our fitted model. Distinguishing a mismatch of models from an error of computation is itself part of the discipline.

The formulation then gives us probabilistic **queries**. These are generally addressed with specific integrals on the probabilistic models: conditioning, marginalization, and summarizing are three operations developed explicitly that may deliver the quantities required to evaluate or optimize performance. They are tractable in some models, but the course deals primarily with how queries can be approached when computation is a challenge.

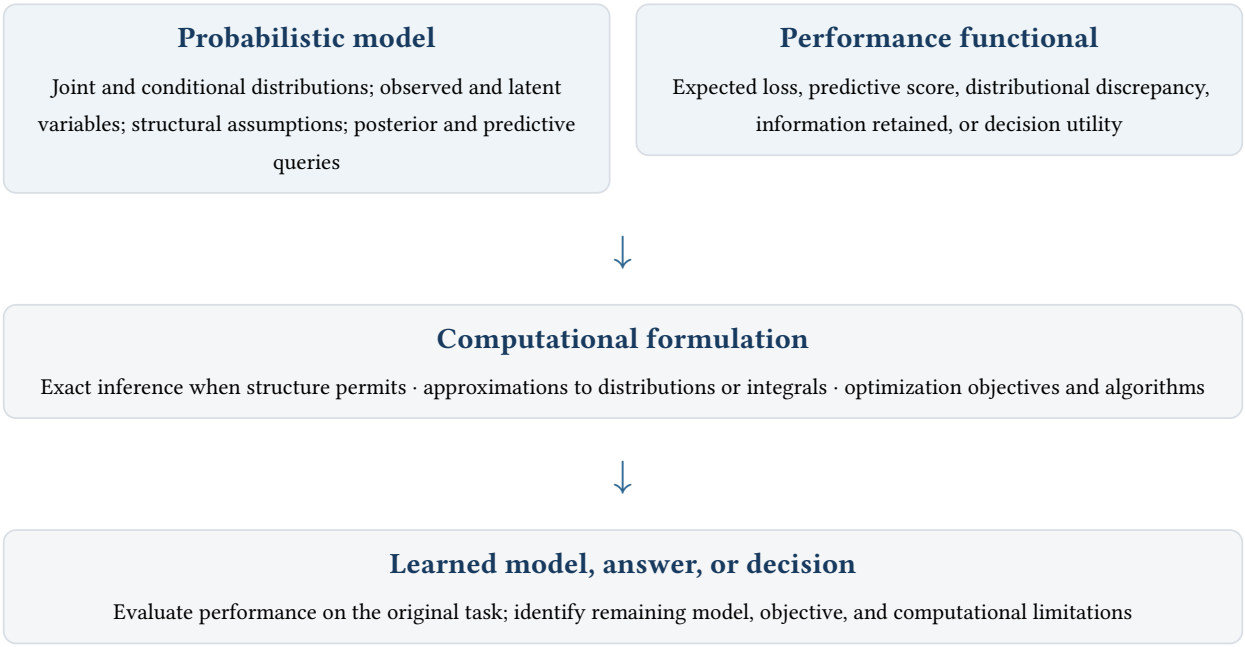
The task–model–performance–computation relationship

Learning problem

Task to be accomplished · available experience and data structure

↓

Mathematical formulation



The arrows describe the logical dependence, not a one-way workflow. Results and computational limits can motivate revising the formulation.

The three major **moves** in the course are consequently extensions of **(i) what models can represent, (ii) what performance functionals can express, and (iii) what computation can achieve**. A task family motivates one move or a combination; no move is an end in itself.

The ideal quantity and the practical calculation

One key move in the course is to substitue a tractable proxy for an ideal mathematical target. A desired performance functional may be replaced during training by an empirical score or surrogate objective. Likewise, an ideal posterior or expectation may be represented by a point estimate, a local Gaussian approximation, a variational distribution, or samples. Some structured problems admit **exact** solutions, so computation is not synonymous with approximation.

The central goal of the course is to develop an understanding of the possible substitutions and the situations in which they are appropriate. Each substitution has different meanings and limitations. The appropriate criterion for assessing the approximation depends on the **original question**, **the probabilistic formulation**, and **the computational power available** not merely whether an internal algorithm converged.

The role of Murphy and mathematical depth

Murphy's books provide the central spine of the course's progression and guide the choice of material, the explanatory logic, the depth of the coverage, and the mathematical language. The course as a whole and each lecture provides only partial coverage, relative to Murphy's second book. Material is selected and occasionally re-ordered.

A detailed understanding of the mathematical development of the material is an essential part of the course. As such, many of the mathematical developments in Murphy are presented in lecture or in the self-work notebooks. The first Murphy book often provides more detailed and scaffolded mathematical derivations, and is used as a reference for the course. The second Murphy book is the primary source fo the underlying logic of the course (as explained above) and for its structure.

How the semester develops

The lecture sequence should be read as a series of responses to **specific kinds of learning problems**, not as a list of algorithms. For every advance we should be able to identify four things: **what task and experience**

motivate it; what probabilistic model and performance question express the task; what prevents the existing mathematics from answering that question satisfactorily; and which new representation, criterion, or computation removes part of that difficulty. The result is a new capability, with assumptions and limitations that matter when we encounter the next problem.

The emphasis may shift among the three moves introduced above: extending the **model**, extending the **performance functional**, or extending the **computation**. They are not mutually exclusive. In particular, changing the practical training objective must not be mistaken for changing the underlying learning task, and choosing a point estimate must not be mistaken for abolishing the posterior. The notebook for each meeting makes the corresponding mathematical advance operational in a particular example; the example does not define the scope of the advance.

Establishing the language of learning problems and models (L01–L03)

L01 — What makes an activity a machine-learning problem? (Framework; exact computation.) We begin by separating the **task** (prediction, generation, discovery, or action), the **experience** available to the learner, and the **performance** by which an answer will be assessed. A dataset does not determine which question is being asked. The probabilistic formulation specifies observed and unknown quantities and a joint distribution; the query identifies what must be conditioned on, marginalized out, or summarized to make a prediction or decision. Linear-Gaussian systems give an unusually favorable baseline: a model of measurements and hidden quantities supports several distinct tasks, and the requisite probability operations can be worked out analytically. The achievement is the common language of **task → formulation → query → answer**, not a general claim that future models will remain Gaussian or computationally easy.

L02 — How can we learn from different observation types without inventing a new theory for each? (Model representation; analytical parameter learning.) Binary observations, continuous measurements, counts, and categorical transitions require different probability distributions, yet many share the exponential-family form. The **model move** is a common parameterization through sufficient statistics, natural parameters, and a log-partition function; its derivatives provide means and covariances. The **computational move** is that maximizing a likelihood can reduce to matching empirical and model expectations of sufficient statistics, rather than using a separate fitting recipe for every distribution. Maximum entropy supplies a complementary characterization when constraints on expected statistics are given. Learning categorical transition probabilities from observed sequences, then generating new sequences, shows how that same machinery answers a concrete generation task. The representation unifies an important family of models, not every possible likelihood.

L03 — How can a model express a large collection of interdependent variables? (Model structure; consequences for possible queries.) Sequences, networks, and spatial observations create joint distributions too large to specify as unrestricted probability tables. The **model move** is factorization: directed conditional distributions or undirected local potentials encode assumptions about how variables interact. Conditional independence, d-separation, and undirected separation say which probabilistic relationships follow from those assumptions; the two graph languages are not interchangeable, and an arrow does not by itself assert causation. Such structure can reduce the parameters needed and later permit efficient queries, but factorization alone does not guarantee easy inference. The Ising/image notebook makes the distinction tangible: it builds a spatial prior and observation likelihood, derives a posterior local conditional, and uses a local maximization procedure. That procedure is **not** a general exact marginalization or a guaranteed globally optimal restoration. L08 will return to the same structural idea when the factorization does make an otherwise costly query exactly tractable.

What counts as success, and how do we fit a model? (L04–L06)

L04 — What if performance concerns a distribution or a particular statistical relationship, rather than a single answer? (Performance functionals and their computational proxies.) Some tasks ask whether a learned distribution resembles the data-generating distribution; others ask whether a representation preserves information about a particular variable, even if it discards much else. Entropy, cross-entropy, and KL divergence let us quantify distributional uncertainty and directed discrepancy. Crucially, expected negative log score equals the target entropy plus KL divergence, so empirical negative log likelihood is a **computable estimate of a principled population criterion**; it is not the population criterion itself. Mutual information and the data-processing inequality provide another kind of performance question: **which dependence or information about a chosen variable survives a representation or processing step?** Neither high mutual information nor low log loss automatically implies low decision loss for every task. The selected short noisy-channel/parity example is a mathematically transparent illustration of distinct criteria—information retained, message error, and resources expended—not the reason machine learning needs these concepts. Three-bit detection is explored in lecture and five-bit correction in the notebook. The broader achievement is the ability to **formulate** and **operationalize** performance questions that recur in prediction, inference, and representation learning.

L05 — How do we predict binary, count, or multiclass outcomes, and learn a useful predictor efficiently? (Model family and gradient-based computation.) The task is supervised probabilistic prediction from observed input–response pairs. A Gaussian regression likelihood is an unsuitable model for Bernoulli labels, counts, or categorical outcomes even if its hierarchical prior is perfectly legitimate. The **model move** is the generalized linear model: combine a predictor with an outcome-appropriate exponential-family observation law and link, including softmax for several classes. L04 now supplies a reason to judge probabilistic predictions with log score; the corresponding empirical likelihood, optionally with a prior penalty, supplies a practical training objective. The **computational move** is to derive and obtain gradients through automatic differentiation and optimize them with stochastic updates. MLE or MAP provides a **point fit**, not a full posterior or a guarantee of good held-out task performance. The Iris notebook checks both the probability model and the parameter updates. The resulting differentiation and training machinery becomes reusable in L09, L11, and L12.

L06 — How can we fit a model when relevant variables are missing or latent? (Computational treatment of marginal likelihood.) With partly observed measurements or unobserved component assignments, the probabilistic model remains a joint distribution over observed and latent quantities. The **learning objective** for maximum-likelihood point fitting concerns the probability of the **observed** data, which requires summing or integrating away the latent values. Filling in a single guess changes that objective; differentiating the marginalized likelihood directly may be inconvenient. The **computational move** in expectation–maximization is to use the conditional distribution of latent quantities at current parameters to construct an expected complete-data objective, then improve the parameter estimate. The lower-bound/KL identity explains the monotonicity of **exact** EM steps and later becomes central to variational inference. The Gaussian missing-data example reconnects L01 conditioning to L02 moment matching: expected second moments require the conditional covariance, not only an imputed mean. EM is useful only when the needed conditional expectations and parameter updates are manageable. It learns a point parameter estimate, not a Bayesian posterior over those parameters; the notebook also assumes an ignorable missingness mechanism.

Answering posterior questions when computation becomes difficult (L07–L10)

These lectures hold the **probabilistic question** in view while comparing mathematically different responses to computational difficulty. A posterior mode, a fitted approximation to a posterior, a structured exact marginal, and a finite Monte Carlo estimate need not give the same answer or be accurate for the same performance question.

L07 — What do we lose by replacing posterior uncertainty with one best-fit value? (Local distributional approximation.) For Bayesian prediction or uncertainty-sensitive decisions, the desired answer may involve a posterior mean, interval, or predictive integral, not merely an optimized parameter value. Even a simple nonlinear Bernoulli regression can produce a posterior that is awkward to normalize or integrate. The **computational move** is Laplace approximation: use the posterior mode and local curvature to replace the nearby posterior shape by a Gaussian, then compute approximate summaries. The model and ideal posterior query remain the same; a local Gaussian is a **proxy for the distribution**, with query-dependent errors when the true posterior is skewed, multimodal, bounded, or poorly represented near its mode. A checked numerical grid provides another approximate point of comparison, not an automatically exact reference. This lecture distinguishes point fitting, posterior approximation, and judging the final predictive query.

L08 — When can dependence structure make apparently impossible inference exact? (Exact computational exploitation of model factorization.) In hidden-state sequences, the task can be to infer the current state from observations so far, reconstruct earlier states using the full record, compute the probability of a sequence, or select the most likely **whole** hidden path. These are different queries and may support different decisions. The model is a Markov chain of hidden states with conditional observation distributions—the structured joint from L02 and L03. Naively summing over every hidden history grows exponentially with sequence length. The **computational move** is to reuse partial sums in forward and backward messages, making filtering, smoothing, and likelihood evaluation **exact** for the specified HMM with known parameters, at a cost proportional to sequence length and quadratic in state count for dense transitions. A path-MAP decision uses a different recursion from state marginals. A short Kalman-filter connection recalls the linear-Gaussian case. This is a success of model structure, not another posterior approximation or a method for learning HMM parameters.

L09 — What if exact posterior calculation is unavailable, but a simpler distribution can be optimized? (Variational approximation and stochastic objectives.) Many tasks require expectations or predictions under a posterior that cannot be computed in closed form. Instead of approximating only its mode, choose a tractable family of distributions and fit one member to the posterior. The **computational move** is variational inference: reuse L06′s evidence lower bound and optimize a restricted approximation by minimizing a directed KL divergence. The lower bound is an **optimization objective for a computational proxy**, not the original downstream performance functional. Mean-field factorization can discard posterior correlations; even a fully optimized approximation may give inaccurate variances for particular linear combinations or inaccurate predictive integrals. Reparameterizing a random draw makes gradients of the stochastic objective accessible with the differentiation machinery from L05. The notebook separates **restriction error** from **optimization error** by comparing a fitted diagonal Gaussian to both an exact regression posterior and the best diagonal approximation.

L10 — How can we estimate a posterior integral without committing to a restrictive approximate family? (Sampling-based numerical approximation.) The mathematical task is often to evaluate an expectation, probability, or decision risk under a specified distribution. Monte Carlo approximates **that integral** by a sample average rather than first replacing the target with a chosen Gaussian or independent-factor family. For difficult posteriors, the **computational move** is to construct a Markov transition whose invariant distribution is the desired target; Metropolis–Hastings corrects proposed moves with an acceptance rule. The distinction is between **correct invariant target** and **accurate answer from a finite run**: mixing, initialization,

autocorrelation, and effective sample size determine the reliability of each estimated quantity, and multimodal targets can expose serious failures. The notebook uses a known two-peak target to assess estimated summaries. This method adds flexibility in the form of numerical sampling error; stationarity alone never certifies a short run.

Richer predictions and learned latent representations (L11–L12)

L11 — What if the conditional relationship and its observation noise are nonlinear? (Model expressivity with familiar fitting and scoring.) For supervised prediction, a linear mean and constant observation variance may both fail to describe the distribution of responses at a given input. The **model move** is to use a small neural network as a differentiable function inside a conditional probability model, predicting an input-dependent Gaussian mean and positive scale. It is the conditional **distribution**, not an unconstrained neural output, that makes probabilistic prediction and log scoring meaningful. The **computational move** largely reuses L05′s automatic differentiation and optimization, now applied to composed nonlinear functions; a Gaussian weight prior connects to penalized point fitting. Compare held-out log score and squared error because they assess different aspects of success. Crucially, the predicted spread describes **observation noise conditional on fitted weights**, not uncertainty from a posterior over the weights. The same expressive maps can later parameterize approximate inference as well as observation models.

L12 — How can we generate complicated observations and infer useful hidden descriptions of them? (Latent generative model, amortized inference, and a representation tradeoff.) Generation and discovery are distinct tasks on the same high-dimensional dataset. A latent generative joint uses a simple prior for hidden coordinates and a neural decoder for the observation distribution; generation draws from that model, whereas discovery asks for the hidden-coordinate posterior given an observed case. The **model move** draws on L03′s latent joint structure and L11′s expressive conditional maps. Exact latent inference is generally inconvenient, so the **computational move** is an encoder network that **amortizes** an approximate posterior across many observations; L06′s bound and L09′s pathwise gradients make joint encoder–decoder training possible. The **performance move** makes the tension between reconstruction quality and a regularizing rate explicit, drawing on L04′s distributional scores and information measures. The average prior KL contains mutual information plus an aggregate-posterior mismatch term: it is **not** identical to information preserved. Posterior collapse and nonidentifiable latent coordinates limit the scientific interpretation of discovered patterns. The notebook assesses the reconstruction/rate tradeoff on multidimensional observations without presenting latent features as identified causes.

L13 — Can students justify the choices they made? (Integration.) Project presentations return to the L01 starting point. Students should state the task, experience and performance criterion; defend the probabilistic model and the relevant query; explain why their optimization or inference procedure is an appropriate computation; and separate limitations of the data, the model, the practical objective, and the numerical method. That synthesis, not the accumulation of algorithms, is the course's intended outcome.

Lecture allocation and notebook commitments

The table below retains the selected 13-meeting allocation and source readings. It is a **planning contract**: L01 is delivered, L02 is agreed, L03 preserves its instructor-approved scope, and L04′s live/notebook split is selected. Other lecture scopes remain proposed pending their own reviews. Each notebook should extend rather than reproduce the live mathematics, normally within a separate hour. Full mathematical source-to-lecture ownership and unresolved feasibility questions remain in the architecture audit and the scope register.

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook
1	**Probabilistic ML and Gaussian systems** Delivered	Machine-learning task framing; probabilistic models; Bayes review; covariance and correlation; multivariate Gaussians; conditioning; missing-data inference; linear Gaussian systems.	• 2.1.5–2.1.6 Conditional probability; Bayes • 2.2.1.1 Bernoulli and binomial • 2.2.4.1 Beta distribution • 2.3.1.1 Gaussian joint distribution • 2.3.1.3–2.3.1.5 Gaussian conditionals; information form • 2.3.2.1–2.3.2.2 Linear-Gaussian systems • 3.2.1 Coin-flipping Bayes • 3.4–3.4.1 Conjugate priors; binomial	• 1.1–1.5 ML tasks and datasets • 2.3 Bayes' rule • 2.4.1 Bernoulli; binomial • 2.7.4 Beta distribution • 3.1.1–3.1.3 Covariance; correlation • 3.2.1–3.3.4 Multivariate and linear Gaussians • 4.3–4.3.1 Empirical risk • 4.6.1–4.6.2 Beta-binomial Bayes	**Gaussian** **delivered** observed uncertaint
2	**Exponential families and learning from sufficient statistics** Agreed scope	Exponential-family representation; Gaussian canonical form; derivatives of the log-partition function; natural and mean parameters; MLE as moment matching; maximum entropy; Gamma fit; categorical counts; Markov-chain learning.	• 2.2.1.2 Categorical distribution • 2.4.1–2.4.2 Exponential-family form; examples • 2.4.3–2.4.4 Log partition; moments • 2.4.5 MLE; moment matching • 2.4.7 Maximum entropy • 2.6.1–2.6.3 Markov chains; language models		**Baby-nam** transitions matrix, co Reference
3	**Graphical models: factorization and conditional independence** Agreed scope restored	Derive DAG factorization from the probability chain rule and independence assumptions; work the student network and parameter economy. Develop d-separation and directed Markov blankets. Define undirected potentials and global normalization, compare separation and independence expressivity, and interpret the binary Ising model. Image posterior and algorithm derivations belong to the notebook.	• 4.2.1–4.2.2 DAG factorization; student network • 4.2.4 / selected 4.2.8 Independence; plates • 4.3.1–4.3.2.1 / 4.3.6.1 Undirected models; Ising; separation • 4.5.1 Directed/undirected independence comparison • 4.3.8 Image restoration: notebook	• Optional 3.6.1–3.6.2 Graphical-model orientation	Construct likelihood conditiona maximizat pixelwise optimum restoration update, an optimum source. **Li** method. It promise th All course
4	**Information theory: mutual information, processing and noisy coding** Proposed teaching scope	Finite entropy, cross-entropy and KL divergence; derive mutual information and sketch the data-processing inequality with its conditional-independence premise. State source-length bounds. Work three-bit parity detection and posterior decoding, distinguishing information, Bayes message risk and a coding guarantee.	• 5.1.3 / 5.1.6; 5.2.1 / 5.2.4 Information and uncertainty • 5.3.1–5.3.4 Shared information and processing • 5.4.1 Code-length bounds: statement only • 5.5 Noisy-channel example: three-bit parity model	• Selected 6.1 / 6.2.1–6.2.2 / 6.3 Uncertainty and shared information	Extend the correcting minimum bit flip; us compute and compa benchmar informatio distributio and guara The lectur notebook Guarantee notebook untested.
5	**Generalized linear models, automatic differentiation and SGD** Proposed teaching scope	Start from the familiar Bayesian regression likelihood and parameter prior; explain why Gaussian observations are unsuitable for binary responses and counts, without implying that the hierarchical model fails. Generalize the observation family using Gaussian/Bernoulli/Poisson GLMs and links. Contrast full posterior inference with MLE/MAP point fitting; derive the canonical NLL and vector softmax gradients, then reverse-mode autodiff and minibatch SGD.	• 15.1 / 15.1.1 / 15.1.3 Common model structure and derivative • 15.1.2 Brief alternative-link comparison • 15.3.2 Three-class probabilities • Selected 6.2; 6.3.1–6.3.2 Automatic differentiation and batch updates	• Selected 7.8.2 / 7.8.5 Matrix derivatives • Selected 8.2 / 8.4.1 / 8.4.3 Gradient and batch-update support	Fit the thr the manua parameter log loss/pr scaffolding derivative a checked Inspected meeting ti the lecture course ada
6	**Latent-variable learning and expectation-maximization** Proposed teaching scope	Recall Bayesian latent-variable hierarchies and joint posterior inference, then distinguish point estimation by marginal likelihood from inference over all unknowns. Show why unknown latent values complicate direct fitting, and derive the ELBO/KL identity, exact E-step tightness and EM monotonicity. Apply Gaussian conditional first/second moments to unknown mean/covariance with missing data; briefly interpret mixture responsibilities and assumptions.	• Selected 6.5.1 / 6.5.3 Probability bound and EM • 6.5.4 Gaussian missing-data updates • 3.11 When the missing-data mechanism can be ignored • 28.2.1 Group-assignment probabilities: brief example	• Selected 3.5 / 8.7 Grouped observations and EM explanation	Assemble conditiona a supplied missing-e **Skill:** Use estimation guarantee updates. I candidate probabiliti fit is not r
7	**Bayesian computation: grid reference and Laplace**	Posterior queries and parameterization-dependent MAP; derive scalar Laplace curvature and normalizer under positive-curvature conditions. Apply to proper-prior	• 7.1–7.3 What posterior questions require: short introduction		Build a sta observatio separately prediction

No.	Lecture	Topics covered	Book 2 — primary	Book 1 — supplemental	Notebook
	approximation Proposed teaching scope	logistic regression, check quadrature domain and resolution, and compare posterior moments and predictive integrals.	• Selected 7.4.1–7.4.3 Most likely parameter, grid and Gaussian shortcut • Selected 7.5 Checking approximation quality • Selected 15.3.5–15.3.6 One-parameter logistic example		the numer approxima from globa **Limit:** Th finite grid exact answ
8	**Exact inference in structured sequence models** Proposed teaching scope	Derive scaled HMM forward/backward recursions, evidence and filtered/smoothed marginals; establish dense computational cost and distinguish joint path MAP from marginal modes. Give a compact Kalman-filter application of L01 Gaussian conditioning, without a second full sequence implementation.	• 9.2.1–9.2.5 Hidden-state calculations and scaling • 9.2.6 Whole history versus separate state decisions • Selected 8.1.2 / 8.2.2 Gaussian filtering: short application		Implemen marginals and an asy errors. Sin supplied. S to obtain e controllin The main previously not includ
9	**Variational inference: posterior geometry and pathwise gradients** Proposed teaching scope	Reuse L06 ELBO identity; optimize restricted reverse KL and work the correlated-Gaussian diagonal optimum. Derive the total pathwise gradient including direct parameter terms and regularity conditions; apply bounded coordinate-ascent VI and recognize amortization.	• 10.1.1–10.1.2 Fitting a restricted distribution • 6.3.3 / 6.3.5; 10.2.1.1 Derivatives through Gaussian draws • 6.3.4 Another gradient estimator: statement only • 10.3.1 Coordinate updates; 10.1.5 learned inference: introduction only		Fit a diago regression Compare solution; u checks and Separate c and differe correct ful **Limit:** Us economic- replaced; r adaptation
10	**Monte Carlo and MCMC: invariance, mixing and estimator error** Proposed teaching scope	Derive IID estimator variance, stationarity/ detailed balance and the general Metropolis–Hastings ratio, retaining rejected states. Explain Gibbs cancellation briefly and dependent-sample error under its conditions. Distinguish invariant transitions, finite-run mixing and function-specific effective sample size; recognize HMC mechanics only.	• 11.2–11.2.2 Averages and sampling error • 2.6.4; 12.2.1–12.2.2 Invariant distribution and acceptance rule • 12.3.1–12.3.2 Conditional-coordinate sampling: brief derivation • Selected 12.6.3 Dependent-sample error; 12.5.3 Hamiltonian mechanics only		Implemen target. Co assess the references query-dep separately for a speci **Limit:** He explained derivative implemen
11	**Neural conditional models: heteroskedastic likelihood and regularization** Proposed teaching scope	Nonlinear feedforward mean/positive-scale model; derive heteroskedastic Gaussian NLL and output derivatives, reusing AD/SGD. Distinguish conditional observation noise from posterior weight uncertainty via total variance. Work the Gaussian-prior/L2 coefficient under mean-loss normalization.	• Selected 16.2.1–16.2.2 / 16.3.1 Small feedforward networks • 15.2.2.1 Gaussian prior and weight shrinkage • 14.2.1 Evaluating predicted probabilities	• 13.2.4.4 Noise that depends on input • 13.5.2 Weight shrinkage; 13.5.1 stopping training: brief mention	Fit one sha held-out N supplied fi law. Match training da conditiona variance d predictive describe o They do n One CPU course ada
12	**Variational autoencoders: amortized inference and rate–distortion** Proposed teaching scope	Generative model versus amortized posterior; joint encoder/decoder objective with event-summed Gaussian NLL and diagonal KL, then observation averaging. Reuse ELBO/pathwise/neural results. Derive rate as mutual information plus aggregate-posterior mismatch; interpret beta, collapse and nonidentifiability.	• Selected 21.1–21.2 Model, inference network and training objective • 21.3.1 Changing the penalty weight • Selected 5.4.2 Reconstruction/ information tradeoff • Selected 21.4 When the model ignores hidden coordinates		Complete joint fit. U comparisc average pr separate. S with a neu rate tradeo or latent p source. **Li** coordinate runtime re identified unrun.
13	**Project presentations and course synthesis**	Student project presentations connecting probabilistic assumptions, computation, prediction, evaluation, uncertainty and scientific interpretation. No additional substantive lecture topic is committed at this stage.			No notebo

Deliberate scope boundaries

This course develops selected mathematical capabilities, not an exhaustive survey of Murphy. The exclusions below are deliberate; they are not hidden promises to cover material during L13 project presentations. The independent architecture audit evaluates source coverage and teaching feasibility.

Other possible topic	Choice in this plan	What we give up
More advanced optimization	Basic derivatives and small-batch updates are sufficient here	No separate study of curved-geometry updates, expensive black-box search, transport or subset selection.
Particle methods for changing hidden states	Exact sequence calculations in L08 and sampling theory in L10	No method for tracking a complex changing state with a cloud of weighted samples (Book 2 Ch. 13).
MRI classification and continuous forecasting	Missing-data fitting and discrete hidden-state lab selected	Less biomedical and forecasting practice. Either application can replace its corresponding lab after review.
More complex regression and prediction checks	Common models in L05 and one neural probability model in L11	No full sparse/hierarchical regression, calibration or guaranteed prediction-interval course.
Larger inference toolkits	One checked method per main lab	No general-purpose graph solver, broad variational-family catalogue or full Hamiltonian sampler derivation.
Bayesian neural networks and Gaussian processes	Small fitted networks only	No separate treatment of uncertainty about network weights or Gaussian-process prediction.
Other generative models	A variational autoencoder in L12	No flow, diffusion or adversarial model teaching. These are different ways to learn data distributions and need their own time.